



A wood engraving of a magpie from the first volume of Thomas Bewick's *History of British Birds*.

CARBON EXISTS IN THREE FORMS

—as graphite, diamond, and charcoal. It was only in the late 18th century, however, that it was established that they are all the same substance. In 1772, French chemist **Antoine Lavoisier** had burned graphite in oxygen and proved that the only product was carbon dioxide. In 1797, British chemist **Smithson Tennant** (1761–1851) repeated the experiment, using diamond instead of graphite. The product was carbon dioxide once again, proving that diamond is an allotrope (form) of carbon.

In this year, amid the turmoil of post-revolutionary France, Italian–French mathematician **Joseph Louis Lagrange** (1736–1813) published *Théorie des Fonctions Analytiques* (*Theory of Analytic Functions*), setting

out a new approach to **calculus**. Although not widely appreciated at the time, his ideas proved invaluable in 20th-century science, especially in the study of **quantum mechanics**.

Thomas Jefferson (1743–1826), vice-president of the US at the time, presented a paper to the American Philosophical Society, Philadelphia, in which he described the fossil remains of a creature he named *Megalonyx* (giant claw). The fossil was that of the **extinct ground sloth**, now known as *Megalonyx jeffersonii*.

British artist and ornithologist **Thomas Bewick** (1753–1828) published the first volume of his *History of British Birds*. He illustrated these volumes using the wood-engraving technique—an advancement in the printing of illustrated books.

760 POUNDS THE WEIGHT OF ROSETTA STONE

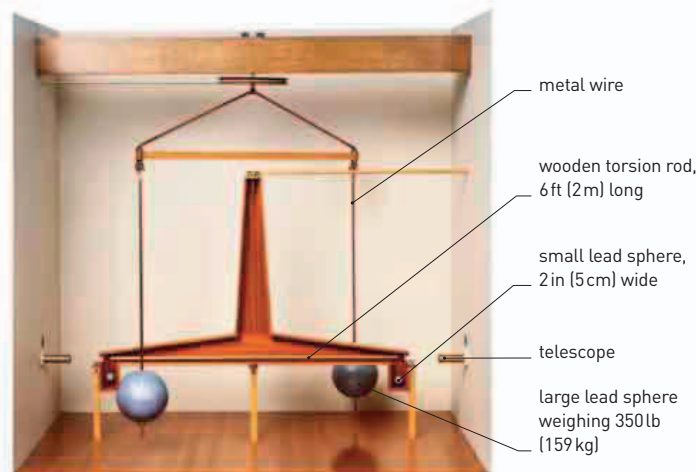


The Rosetta Stone is a decree passed on behalf of Egyptian King Ptolemy V in 196 BCE. It carries the same text in three scripts.

FRENCH PHARMACIST AND CHEMIST LOUIS VAUQUELIN

(1763–1829) discovered **beryllium**, a metal that melts at 2,349°F (1,287°C), in 1798. He used a process of chemical extraction to obtain beryllium from a variety of emerald known as beryl.

British economist and demographer **Thomas Robert Malthus** (1766–1834), then an Anglican priest in Surrey, England, anonymously published the first edition of *An Essay on the Principle of Population*. The book argued that **population growth** would ultimately outstrip the carrying capacity of Earth. It had a profound influence on Charles Darwin in his development of the idea of a struggle for survival (see pp.204–05).



Cavendish torsion balance

This model of Cavendish's torsion balance experiment shows the arrangement of small and large lead spheres attached to a wooden rod.

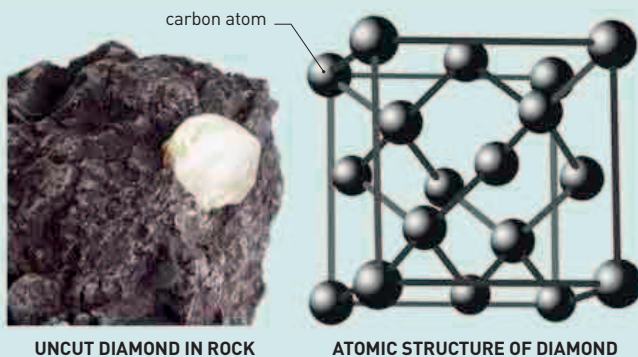
One of the most important experiments in the history of science was completed in 1798 by reclusive British physicist **Henry Cavendish** (1731–1810). He used a **torsion balance** apparatus, designed by British geologist and astronomer John Michell (1724–93), to measure the mass of Earth. The apparatus had a wooden rod suspended from a wire, with a lead sphere attached to each end, and two smaller lead spheres. He measured the force or **gravitational attraction** between the large and small lead spheres, and calculated the gravitational force of Earth on

the small ball by weighing it. The ratio of these two forces allowed him to **calculate the mass and density of Earth**. He found the density to be 5.48 times that of water—the modern estimate is 5.52 times. Cavendish's careful calculation of possible errors and the painstaking detail of his work arguably make this the first modern physics experiment.

Meticulous experimentation was also a feature of the work of American-born British physicist **Benjamin Thompson** (1753–1814). While working in Bavaria, Germany, he investigated the way **heat was produced by friction** when cannons were being bored, and disproved the idea that heat was a kind of fluid, called caloric. Thompson published these findings in a paper in 1798.

CRYSTALLINE CARBON

A crystalline form of carbon, diamonds are formed at high pressure within the Earth and brought to the surface by tectonic activity. They are found in up to 3-billion-year-old rock strata. Carbon atoms are arranged in a crystal structure called a diamond lattice (a face-centered cube). This rigid structure is why diamonds are hard and also transparent.



UNCUT DIAMOND IN ROCK

ATOMIC STRUCTURE OF DIAMOND

Smithson Tennant proves that diamond is a pure form of **carbon**

Joseph Louis Lagrange's *Theory of Analytic Functions* is published

Thomas Jefferson identifies *Megalonyx*

Thomas Bewick publishes the first volume of his *History of British Birds*

French chemist Louis Vauquelin discovers **chromium**

1798 Louis Vauquelin discovers **beryllium**

1798 Caroline Herschel publishes updated version of Flamsteed's star catalogue

1798 Joseph Louis Lagrange publishes *Résolution des Équations Numériques*

1798 Henry Cavendish measures the **mass of Earth**

1799 Pierre-Simon Laplace begins publication of *Traité de Mécanique Céleste*

1798 Thomas Robert Malthus publishes *An Essay on the Principle of Population*

1798 Edward Jenner publishes a paper on heat and friction

1798 Benjamin Thompson publishes *An Experimental Enquiry Concerning the Source of the Heat which is Excited by Friction*

July 1799 Pierre Bouchard finds the **Rosetta Stone**

1799 Alexander von Humboldt and Aimé Bonpland begin exploration of the **natural history of South America**